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Sub : JAVA PROGRAMMING
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Q1) write a simple program demonstrating the servlet. life cycle.

Servlet Lifecycle :

init() → called once when the Servlet is first loaded.
doGet() → called for every request.
destroy() → called once before the Servlet is unloaded.

code : (lifecycle.java).

```
import java.io.Exception;
import java.io.PrintWritter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.webservlet;
@webservlet("/lifecyle"); (no need web.xml)
public class lifecycle extends HttpServlet {
    @override
    public void init() throws ServletException {
        system.out.println("init called.");
    }

    @override
    protected void doGet(
                    HttpservletRequest request,
                    HttpservletResponse response.
                ) throw servletException, IoException {
      response.setContentType("text/html");
      printwritter out = response.getWritter();
```

```
        out.println("<html><body>");
        out.println("<h3>servlet life cycle demo</h3>");
        out.println("<p>Request processed</p>");
        out.println("</body></html>");
    }

    @override
    public void destroy() {
        System.out.println("servlet destroyed");
    }
}
```

execution Steps:

1 → javac -cp "servlet-api.jar" lifecycle.java

2 → move to the compiled class (lifecycle.class) into. tomcat xxx /webapps/appname/WEB-INF/classes.

3 → startup.bat (tomcat server).

4 → browser ://. (http://localhost:8080/appname/lifecycle

output:

localhost:8080 .... – ☐ ×

server Life cycle demo.

Request processed

(Q2) To create a Helloworld Java servlet program.

code: (helloworld.java).

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.webservlet;

@webservlet("/helloworld");
public class helloworld extends HttpServlet {
    @override
    public void doGet(HttpServletRequest request, HttpServlet
                Response response) throws ServletException,
                IOException {
        repone.setContentType("text/html");
        printwriter out = response.getwriter();
        out.println("<html><body>");
        out.println("<h3>Hello Udhayarajan J</h3>");
        out.println("</body></html>");
    }
}
```

execution:

1 → javac -cp "servlet-api.jar" helloworld.java.

2 → move to the compiled class into (tomcatxx/hello/WEB-INF/classes).

3 → startup.bat.

4 → browser // (http://localhost:8080/hello/helloworld).

Output:

localhost 8080/hell

Hello Vidhyarajan !

Q5) Creating simple JSP page for login.

Code:

login.jsp:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Login page</title>
</head>
<body>
    <h2>simple Login page</h2>.
    <form action="login" method="post">
        Username : <input type="text" name="username" />
        Password : <input type="password" name="password" />
        <input type="submit" value="Login" />
    </form>
</body>
</html>.
```

login.java.

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/login")
public class login extende HttpServlet {
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String uger = request.getParameter("username");
        String pase = request.getParameter("password");
        if(uger.equals("admin") && pase.equals("123")){
            out.println("<h2> Login Successfully </h2>");
        }
        else {
            out.println("<h2> Invalid Login </h2>");
        }
    }
}
```

Structure:

```
appname
  └ lib
  │   └ servlet-api.jar.
  └ src
  │   └ login.java.
  └ WEB-INF/classes
  │              └ login.class.
  └ login.jsp.
```

Page No:
Date:

executation:

1 → javac -cp ".;lib/servlet-api.jar" -d "WEB-INF/classes" src/*.java

2 → browser :// (http://localhost:8080/Login/Login.jsp

output:

①

login page — ☐ ×

Simple Login page

Username : [admin]

password : [123]

[Login]

②.

localhost:8080/.... — ☐ ×

Login successfully!.

Q4) Creating a Simple JSP page for Addition & multiplication.

Code:

calc.jsp.

```
<!DOCTYPE html>
<html>
<head>
    <title> Calculation </title>
</head>
<body>
    <form action="calc" method="POST">.
        Number 1: <input type="number" name="num1" />
        Number 2: <input type="number" name="num2" />
        <input type="Submit" value="Calculate" />.

        <% if(request.getAttribute("Add") != null &&
                request.getAttribute("mul") != null) { %>
            <h2> Result </h2>
            Addition: ${add} </br>
            Multiplication: ${mul}
        <% } %>.
</body>
</html>.
```

calc java.

```
import java.io.*;
import jakarta.servlet.*;
import jarkata.servlet.http *;
import jarkata.servlet.annotation.webservlet;

@webservlet("/calc");
public class calc extends Httpservlet {.
    @override
    protected void doPost(HttpServletrequest request,
                HttpServletresponse response)
        throws IOException, ServletException {.
```

```
int a = Integer.parseInt(request.getParameter("num1"));
int b = Integer.parseInt(request.getParameter("num2"));

int add = a+b;
int mul = a*b;

request.setAttribute("add", add);
request.setAttribute("mul", mul);

request.getRequestDispatcher("calc.jsp").forward
                        (request, response);
    }
}
```

output:

calculation - □ ×

Number 1 : 12

Number 2 : 12

Calculate

Result :

Addition : 24

Multiplication : 144

Q8). Creating a simple JSP page for Student registration.

Code:

register.jsp

```
<!DOCTYPE html>
<Html>
<head>
  <title> Student registration </title>
</head>
<body>
   <form action="register" action="post">
        Name: <input type="text" name="name" />
        Age : <input type="text" name=" Age" />
        Email : <input type="email" name="email" />
        Course: <input type="text" name="Course" />
        <input type="submit" value="Register" />
   </form>
</body>
</html>
```

register.java.

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.webservlet;

@webservlet("/register")
public class Register extends Httpservlet {
     @override
     protected void doPost(HttpServletrequest request,
                       HttpServletresponse response) throws
                  IOException, ServletException {
```

```
        string name = request.getParameter("name");
        string Age = request.getParameter("Age");
        string email = request.getParameter("email");
        string course = request.getParameter("course");

        request.setAttribute("name", name);
        request.setAttribute("age", Age);
        request.setAttribute("email", email);
        request.setAttribute("course", course);

        request.getRequestDispatcher("success.jsp").
                        forward(request, response);
    }
}
```

Success.jsp.

```
<!DOCTYPE html>
<html>
<head>
  <title> Registration successfully </title>
</head>
<body>
    <h2> student Registered successfully </h2>.
    <p> <b>Name </b> ${name} </p>
    <p> <b> Age </b> ${age} </p>
    <p> <b> Email </b> ${email} </p>
    <p> <b> Course </b> ${course} </p>
</body>
</html>.
```

output:

①

Student Registration

- ☐ ×

Name : xyz

Age : 10

Email : xyz...

course : MCA

Register

②.

Registration successfully

- ☐ ×

student registered successfully!

Name : xyz

Age : 10.

Email : xyz@email.com

course : MCA

Q6) write a program to connect JSP application. employee application form with MySQL.

table: employee (name, email, salary)

database: dbms_2202561

code:

employee.jsp.

```
<!DOCTYPE html>.
<html>
<head>
    <title> Employee Registration </title>.
```

```
</head>
<body>
    <form action="employee" method="post">
        Employee Name: <input type="text" name="name" />
        email: <input type="email" name="email" />
        Salary: <input type="text name="salary" />
        <input type="submit" value="Register"/>
    </form>
</body>
</html>.
```

employee.java.

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;
import java.sql.*;

@WebServlet("/employee"):
public class employee Extends HttpServlet {
    @override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws IOException, ServletException {
        String name = request.getParameter("name");
        string email = request.getParameter("email");
        String salary = request.getParameter("Salary");
        Connection con=null;
        preparedStatement ps=null;
        try {
            class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/dbms_220256
                "root",
                "ATVCACOO");
```

```
String sql = "Insert into employee(name, email, Salary)
                    values(?, ?, ?)";
ps = con.preparedStatement(sql);
ps.setString(1, name);
ps.setString(2, email);
ps.setString(3, Salary);
ps.executeUpdate();
response.setContentType("text/html");
printWriter out = response.getWriter();
out.println("<h2> Employee registered successfully
            </h2>");
}
catch(Exception e) {
    e.printStackTrace();
}
}
}
```

output :

①

Employee registration — □ ×

Employee Registration Form

employee Name : xyz

Email : xyz@..

Salary : 12000

Register

②

localhost:8080/... — □ ×

Employee Registered successfully!